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CONFORMAL AND SMOOTH TITANIUM NITRIDE LAYERS AND METHODS OF FORMING THE SAME

Abstract

The disclosed technology generally relates to forming a thin film comprising titanium nitride (TiN), and more particularly to forming by a cyclical vapor deposition process the thin film comprising (TiN). In one aspect, a method a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, wherein a ratio of the N precursor flow rate to the Ti precursor flow rate exceeds 3. The method is such that the TiN thin film has a preferential (111) crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4. Aspects are also directed to semiconductor structures incorporating the thin film and method of forming the same.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS [0001] Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57. [0002] This application is a continuation of U.S. application Ser. No. 17/547,086, filed Dec. 9, 2021, entitled “CONFORMAL AND SMOOTH TITANIUM NITRIDE LAYERS AND METHODS OF FORMING THE SAME,” which is a continuation in part of U.S. application Ser. No. 16/595,945, filed Oct. 8, 2019, entitled “CONFORMAL AND SMOOTH TITANIUM NITRIDE LAYERS AND METHODS OF FORMING THE SAME,” and claims the priority benefit under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/123,733, filed Dec. 10, 2020, entitled “CONFORMAL AND SMOOTH TITANIUM NITRIDE LAYERS AND METHODS OF FORMING THE SAME,” the content of each which is hereby expressly incorporated by reference in its entirety.

BACKGROUND

Field

[0003] The disclosed technology generally relates to forming a titanium nitride layer, and more particularly to a conformal and smooth titanium nitride layer.

Description of the Related Art

[0004] Titanium nitride (TiN) has been widely used in fabrication of various structures in integrated circuits (ICs). For example, TiN has been used in diffusion barriers, various electrodes and metallization structures. Such wide usage of TiN in IC fabrication can be attributed to its structural, thermal and electrical properties. As the dimensions of various IC structures shrink, TiN is formed on features having increasingly smaller dimensions and complex topologies. For example, as the technology node scales to 10 nm node and beyond, there is a need for TiN layers, e.g., as diffusion barriers, that can conformally line high aspect ratio trenches and vias having dimensions as small as few nanometers. While techniques such as physical vapor deposition (PVD) and chemical vapor deposition (CVD) have been used in the IC industry to form TiN for decades, the increased need for conformality of TiN films to be deposited in smaller trenches or vias may eventually limit their usage. On the other hand, while atomic layer deposition (ALD) has been demonstrated for conformal deposition of TiN films, some electrical properties (e.g., conductivity) and physical properties (e.g., surface roughness) of the film may be inferior compared to TiN films formed using other methods such as physical vapor deposition (PVD). Thus, there is a need for atomic layer deposition methods for forming TiN-based films with superior surface smoothness and step coverage, while also having matching or superior electrical and physical properties,

relative to TiN films formed by PVD and CVD, for use in IC fabrication.

SUMMARY

[0005] In one aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a first portion of the thin film by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor and an exposure to a first N precursor. The method additionally comprises forming on the first portion of the thin film a second portion of the thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor. The exposures to one or both of a Ti precursor and a N precursor during the one or more second ALD cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more first ALD cycles.

[0006] In another aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises providing a semiconductor substrate comprising a trench or a via having an aspect ratio exceeding 1. The method additionally comprises forming the thin film in the trench or the via by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor and an exposure to a first N precursor to form a first portion of the thin film in the trench or the via. The method additionally comprises exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor to form a second portion of the thin film on the first portion of the thin film. Exposures to one or both of the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles are at different pressures relative to corresponding exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles.

[0007] In another aspect, a semiconductor structure comprises a semiconductor substrate comprising a non-metallic sidewall surface in a trench or a via having an aspect ratio exceeding 5. The semiconductor structure additionally comprises a thin film comprising TiN conformally coating the non-metallic sidewall surface, wherein a ratio of thicknesses of the thin film on formed lower 25% of a height of the trench or the via and upper 25% of the height of the trench or the via exceeds 0.9.

[0008] In another aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, wherein a ratio of the N precursor flow rate to the Ti precursor flow rate exceeds 3. The method is such that the TiN thin film has a preferential (111) crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0009] In another aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, wherein the N precursor flow rate exceeds 500 sccm. The method is such that the TiN thin film has a preferential (111) crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0010] In another aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate. The method additionally comprises forming on the TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate. The method is such that one or both of the TiN thin film and the second TiN thin film have a preferential (111) crystalline texture such that X-ray spectra of the one or both of the TiN thin film and the second TiN thin film have a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0011] In another aspect, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a first TiN thin film at a first pressure by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate. The first TiN thin film has a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4. The method additionally comprises forming on the first TiN thin film a second TiN thin film at a second pressure higher than the first pressure by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate.

[0012] In another aspect, a semiconductor structure comprises a semiconductor substrate comprising a non-metallic sidewall surface in a trench or a via having an aspect ratio exceeding 5. The semiconductor structure additionally comprises a thin film comprising TiN conformally coating the non-metallic sidewall surface, wherein the thin film has a crystalline texture such that an X-ray diffraction profile of the thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to a peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation exceeds 0.4.

[0013] In another aspect, a semiconductor structure comprises a semiconductor substrate comprising a non-metallic sidewall surface in a trench or a via having an aspect ratio exceeding 5. The semiconductor structure additionally comprises a thin film comprising TiN conformally coating the non-metallic sidewall surface, wherein the thin film comprises a lower portion and an upper portion, where the lower and upper portions have different crystalline textures. At least the lower portion of the thin film has a crystalline texture such that an X-ray diffraction profile of the thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to a peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation exceeds 0.4.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] FIGS. 1A-1D schematically illustrate nucleation and growth mechanisms of thin films under different growth modes.

[0015] FIG. 2 is a cross-sectional transmission electron micrograph of a TiN layer grown on an

oxide-coated substrate by thermal atomic layer deposition.

[0016] FIG. 3A is a flow chart schematically illustrating an atomic layer deposition method of forming a TiN layer by exposing a substrate to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0017] FIG. 3B schematically illustrates a cross-sectional view of a semiconductor structure comprising a TiN layer formed by an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0018] FIG. 4 schematically illustrates pressure traces of different cycles of an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0019] FIG. 5 schematically illustrates a cross-sectional view of a via lined with a TiN layer having different thicknesses at different portions of the via.

[0020] FIG. 6 is a graph showing experimentally measured surface roughness and step coverage trends as functions of thickness for TiN layers formed by an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0021] FIG. 7A is a cross-sectional transmission electron micrograph of high aspect ratio vias lined with a TiN layer formed by an atomic layer deposition method in which a substrate is exposed to ALD cycles performed at the same precursor exposure pressure.

[0022] FIG. 7B is a cross-sectional transmission electron micrograph an upper region of the high aspect ratio vias shown in FIG. 7A.

[0023] FIG. 7C is a cross-sectional transmission electron micrograph of a lower region of the high aspect ratio vias shown in FIG. 7A.

[0024] FIG. 8A is a cross-sectional transmission electron micrograph of a TiN layer formed at an upper region of a high aspect ratio via similar to that shown in FIG. 7A by an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0025] FIG. 8B is a cross-sectional transmission electron micrograph of a TiN layer formed at a lower region of the high aspect ratio via trench shown in FIG. 8A.

[0026] FIG. 9 is a graph showing statistical comparison of measured step coverages between a TiN layer formed by atomic layer deposition at a single exposure pressure and a TiN layer formed by atomic layer deposition at a plurality of exposure pressures, according to embodiments.

[0027] FIG. 10 schematically illustrates a cross-sectional view of a via lined with a TiN layer formed by an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

[0028] FIG. 11 is a flow chart schematically illustrating an atomic layer deposition method of forming a TiN layer having high (111) crystalline texture by exposing a substrate to a relatively high N precursor flow rate, according to embodiments.

[0029] FIG. 12A illustrates experimental X-ray diffraction spectra of TiN thin films having varying degrees of (111) crystalline texture, according to embodiments.

[0030] FIG. 12B is a graph illustrating the experimental ratios of a peak height of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN obtained from the X-ray diffraction spectra of FIG. 12A.

[0031] FIG. 13 is a graph of experimental thickness and resistivity measurements of TiN thin films having increased (111) crystalline texture, according to embodiments.

[0032] FIG. 14 is a graph of experimental hardness and modulus measurements of TiN thin films having increased (111) crystalline texture, according to embodiments.

[0033] FIG. 15 is a graph of experimental hardness measurements of TiN thin films having

increased (111) crystalline texture, according to embodiments.

[0034] FIG. 16 illustrates experimental X-ray diffraction spectra of TiN thin films formed at different exposure pressures and having increased (111) crystalline texture, according to embodiments.

[0035] FIG. 17 illustrates chlorine concentration depth profiles of TiN thin films having increased (111) crystalline texture, according to embodiments.

DETAILED DESCRIPTION

[0036] As described above, there is a need in the integrated circuit (IC) industry for smooth and conformal TiN films with superior electrical and physical properties, as well as methods of forming such films. To address these and other needs, disclosed herein is a smooth and conformal thin film comprising TiN and a cyclical vapor deposition method of forming the thin film, which displays the conformality characteristic of a film deposited by cyclical vapor deposition processes, while also having electrical and physical properties that are superior or matching those of TiN films formed by existing physical vapor deposition (PVD) and chemical vapor deposition (CVD) methods. In particular, a method of forming a thin film comprising titanium nitride (TiN) comprises forming on a semiconductor substrate a first portion of the thin film by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor and an exposure to a first N precursor. The method additionally comprises forming on the first portion of the thin film a second portion of the thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor. The exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are different compared to corresponding exposures to one or both of the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles. The cyclical vapor deposition processes disclosed herein are sometimes referred to as atomic layer deposition (ALD). However, cyclical vapor deposition processes are not limited to atomic layer deposition processes. For example, the precursors may partly or substantially saturate a reaction surface in various embodiments described herein.

[0037] By exposing the substrate to the Ti and/or N precursors at relatively low pressures, e.g., less than 3 torr, during deposition of the first portion of the thin film, the initial film growth may proceed substantially in a layer-by-layer growth mode, which advantageously results in an average grain size that is lower and a surface roughness that is lower relative to a comparable TiN film that is deposited by exposing the substrate to the Ti and/or N precursors at higher pressures, e.g., greater than 3 torr or 5 torr. On the other hand, by exposing the substrate to the Ti and/or N precursors at relatively high pressures, e.g., greater than 3 torr, during deposition of the second portion of the thin film, the latter portion of the film growth advantageously results in a degree of conformality or a step coverage that is higher relative to a comparable TiN film that is deposited by exposing the substrate to the Ti and/or N precursors at the relatively low pressures, e.g., less than 3 torr or less than 1 torr.

[0038] In addition, because the first portion of the TiN film grows in a layer-by-layer mode, the second portion of the thin film may continue to grow, using the first portion as a template, in a layer-by-layer mode compared to a comparable thin film grown starting with exposures to the Ti and/or N precursors at relatively higher pressures.

[0039] As a net result, when deposited on certain surfaces, e.g., a surface comprising a non-metallic surface, the thin film comprising the first and second portions deposited by depositing at two different corresponding exposure pressures for one or both of the Ti precursor and the N precursor according to methods disclosed herein advantageously has a combination of surface roughness and conformality that is superior relative to a thin film layer formed on the same surface using a single pressure. Alternatively, or in addition, in part owing to the improved smoothness and conformality, the thin film has a relatively low electrical resistivity compared to TiN layers formed by some

existing methods.

[0040] As described herein, a compound referred to by its constituent elements without specific stoichiometric ratios thereof shall be understood to encompass all possible nonzero concentrations of each element unless explicitly limited. For example, titanium nitride (TiN) shall be understood to encompass all possible stoichiometric and nonstoichiometric compositions of titanium nitride that can be expressed by a general formula $\text{Ti}_{x_1}\text{N}_{x_2}$, where $x_1 > 0$, including TiN, $\text{Ti}_{0.3}\text{N}_{0.4}$, $\text{Ti}_{0.4}\text{N}_{0.3}$, $\text{Ti}_{0.6}\text{N}_{0.5}$, $\text{Ti}_{0.2}\text{N}$ and $\text{TiN}_{0.2}$ as well as other non-stoichiometric compositions of Ti and N.

[0041] As described above, titanium nitride (TiN) plays an important role in integrated circuit (IC) fabrication. While techniques such as physical vapor deposition (PVD) and chemical vapor deposition (CVD) have been used in the IC industry to deposit TiN, the need for deposition methods for forming TiN-based films having high conformality without significant compromise in electrical and physical properties has been increasing.

[0042] In addition, while plasma-enhanced processes such as plasma enhanced atomic layer deposition (PE-ALD) may be effective in forming conformal films on surfaces having relatively low aspect ratios, such processes may not be effective in depositing films inside vias and cavities having relative high aspect ratios. Without being limited by theory, one possible reason for this is that a plasma or its active species may not reach deeper portions of high aspect ratio vias under some circumstances. In these circumstances, different portions of vias may be exposed to different amounts of the plasma or its active species, leading to undesirable structural effects of non-uniform deposition, such as thicker films being deposited near the opening of the via compared to deeper portions (sometimes called cusping or keyhole formation). For these reasons, thermal ALD may be more advantageous, because thermal ALD does not depend on the ability of the plasma or its active species to reach portions of the surface being deposited on.

[0043] However, while thermal ALD techniques may be suitable for forming relatively conformal TiN films on topography, particularly topography with relatively high aspect ratios (e.g., over 1:1), the inventors have recognized that TiN films formed by thermal ALD can be inferior to TiN films formed by PVD or CVD in some respects, e.g., film roughness and electrical resistivity. In this regard, the inventors have discovered that some electrical properties and/or physical properties of ALD-grown TiN-based films can be affected by the mode of growth. In particular, the inventors have discovered that, while it may be desirable to grow the TiN-based films in a two-dimensional layer-by-layer growth mode in ALD, such layer-by-layer growth mode may not be easily achieved under some circumstances. The inventors have further discovered that growing TiN-based films by ALD in a layer-by-layer growth mode poses a particular challenge in IC fabrication where the TiN-based films are formed on non-metal surfaces, such as insulating surfaces, such as oxide and nitride surfaces or semiconductor surfaces such as doped and undoped silicon surfaces. The inventors have recognized that the degree to which the TiN-based films may be grown in a layer-by-layer growth mode may in turn be dependent on the initial growth mode that is dependent on the type of surface, as described herein without being bound to any theory, in reference to FIGS. 1A-1D.

[0044] FIG. 1A schematically illustrates nucleation of a TiN layer and FIGS. 1B-1D illustrate different growth modes of the TiN layer on different surfaces. Referring FIG. 1A, once precursor molecules **104** reach the surface of a substrate **100**, they become physically adsorbed thereon. Some of the adsorbed molecules **104** may diffuse along the surface of the substrate **100** until they reach an energetically favorable position to be chemisorbed. The surface diffusion is governed by, among other things, the substrate temperature, the substrate material and kinetic energy of the adsorbed molecules. When the size of a nuclei formed by chemisorbed molecules exceeds a certain size (sometimes referred to as “critical size”) determined by the trade-off between volume free energy and surface energy, the nuclei may become energetically stable, and start to grow in size. Thus formed layer **108** of stable nuclei continue to grow by incorporating additional precursor molecules **104**. Subsequent film growth can be classified according to different growth modes,

schematically illustrated in FIGS. 1B-1D.

[0045] FIG. 1B schematically illustrates a three-dimensional island growth mode, sometimes referred to as Volmer-Weber growth mode, which results in the formation of a layer **112** of three-dimensional islands. Without being bound to any theory, the island growth mode can dominate when the net surface free energy associated with three-dimensional islands is positive, indicating that deposited atoms are more strongly bound to each other than to the substrate. It will be appreciated that the energetics of ALD growth of TiN layers can favor the island growth mode, e.g., when the metallic TiN layers are deposited on some semiconductor and/or insulating material surfaces.

[0046] FIG. 1C illustrates a layer-by-layer growth mode, sometimes referred to as Frank-van der Merwe growth mode, which results in the formation of a relatively smooth two-dimensional layer **116**. Without being bound to any theory, the layer-by-layer growth mode can dominate when the deposited atoms are more strongly bound to the substrate than to each other, such that a stable two-dimensional layer **116** is energetically favored. The layer-by-layer growth mode can be sustained when there is a continuous decrease in bonding energy between the layers from the first monolayer to the bulk-crystal value of the TiN layer.

[0047] While FIGS. 1B and 1C are two different possible growth modes of thin films, it will be appreciated that, under some circumstances, a growth mode that is intermediate between a layer-by-layer growth mode and a three-dimensional growth mode is possible. FIG. 1D illustrates an example of an intermediate growth mode known as Stranski-Krastanov (SK) growth mode. Without being bound to any theory, the SK growth may occur in thin film growth that commences in a layer-by-layer mode. When layer-by-layer growth becomes unfavorable after the formation of one or more monolayers, an island growth mode starts to dominate over a layer-by-layer growth mode, resulting in thin film structure **120** in which three dimensional islands are formed on a two-dimensional initial layer. The SK growth mode can occur as a strain relaxation mechanism (strain-induced roughening).

[0048] In addition to the interaction between the deposit and the substrate, other factors such as the substrate temperature, reactor pressure and deposition rate can significantly affect the nucleation and early growth processes, which in turn affects the final nanostructure or microstructure of the resulting thin film. For example, deposition conditions that enhance surface diffusion, e.g., relatively high substrate temperatures, relatively low pressures and/or low deposition rates may promote the growth in a layer-by-layer mode. Hence, as disclosed herein, by enhancing surface diffusion during deposition of an initial portion of a TiN film by, e.g., lowering pressure and growth rate, the initial film growth according to embodiments may proceed substantially in a layer-by-layer growth mode.

[0049] It has been discovered that, when TiN is grown by ALD on various surfaces of interest in IC fabrication, such as dielectric and semiconductor surfaces, the ALD growth initializes in a three-dimensional island growth mode or a SK growth mode. For example, under some circumstances, ALD growth of TiN on substrate surfaces, including doped and undoped Si, SiO₂, Si₃N₄ and other high K or low K materials, may proceed in an island growth mode or the SK growth mode. The inventors have discovered that, in part owing to the initial growth mode of either an island or SK growth mode, subsequent growth of TiN by ALD often results in a film morphology that is undesirable for various applications of ultrathin conformal TiN for high aspect ratio structures, as illustrated in FIG. 2.

[0050] FIG. 2 is a cross-sectional transmission electron micrograph of a TiN layer grown by thermal ALD on a Si substrate coated with native oxide. After an initial film grown in either a three-dimensional island or SK growth mode, the ALD growth of TiN is often characterized by a competitive growth of adjacent crystals with different orientations, resulting, under some circumstances, in V-shaped grains close to the nucleation layer and culminating in a columnar morphology at higher film thicknesses. As illustrated in FIG. 2, the resulting film morphology

includes faceted column tops that give rise to a significant surface roughness and column boundaries having lower density relative to the grains. It will be appreciated that the column boundaries can have significantly worse diffusion barrier properties relative to the grains themselves, and may serve as paths of least resistance for transportation of undesirable contaminant through the TiN layer.

[0051] The inventors have discovered that, when an initial portion of a TiN layer is formed on the non-metal surface by exposing the substrate to Ti and/or N precursors at relatively low pressures, e.g., less than 1 torr, an initial three-dimensional or SK growth mode can be suppressed and a layer-by-layer growth mode can be promoted in an initial stage, e.g., a nucleation stage, of the TiN deposition. Among other reasons, this may be because the local diffusion of adsorbed Ti and N precursor molecules have more time to locally diffuse and wet the substrate surface, especially a non-metal surface, with relatively low contact angles. The TiN layer grown at relatively low exposure pressures results in a layer that uniformly covers large areas of the non-metal surface without substantially forming islands, such that, the initial stages of growth tend more to favor a layer-by-layer growth mode on substrate surfaces on which ALD TiN would normally favor a three-dimensional island or SK growth mode as described above. Thus, by initiating ALD of TiN by exposing the substrate to the Ti and/or N precursors at relatively low precursor exposure pressures, e.g., less than 3 torr, the resulting initial layer may grow in a layer-by-layer mode, e.g., in the nucleation stage. A subsequent bulk stage of growth, which may proceed by exposing the substrate to the Ti and/or N precursors at the relatively high precursor exposure pressures, e.g., greater than 3 torr, may continue to proceed in a layer-by-layer mode. By employing the methods according to embodiments, some of the drawbacks of conventional ALD of TiN may be avoided, particularly when the TiN layer is formed by ALD directly on some semiconductor and/or insulating materials, particularly inorganic layers comprising Si, SiO₂ and/or Si₃N₄, which may ordinarily be associated with an initial growth characterized by an island or SK growth mode followed by a columnar growth as described above.

[0052] FIG. 3A is a flow chart schematically illustrating an atomic layer deposition method **300** of forming a TiN layer by exposing a substrate to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments. The resulting film may have at least two regions formed at different corresponding exposure pressures. FIG. 3B schematically illustrates a cross-sectional view of a semiconductor structure **350** comprising a TiN layer formed by an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures, according to the method illustrated in FIG. 3A.

Referring to FIG. 3A, the method **300** includes providing **310** a substrate comprising a non-metal surface in a reaction chamber configured for ALD, e.g., thermal ALD. The method **300** additionally comprises an initial stage, e.g., a nucleation stage, that includes forming **320** on the substrate a first portion of the thin film by exposing the semiconductor substrate to one or more first ALD cycles each comprising an exposure to a first Ti precursor and a first N precursor at first respective exposure pressures. The method **300** further comprises latter stage, e.g., a bulk deposition stage, including forming **330** on the first portion of the thin film a second portion of the thin film by exposing the semiconductor substrate to one or more second ALD cycles each comprising an exposure to a second Ti precursor and a second N precursor at second respective exposures. The exposures to one or both of the Ti precursor and the N precursor during the one or more second ALD cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more first ALD cycles.

[0053] Referring to FIG. 3B, a cross-sectional view of a semiconductor thin film structure **350** comprising a substrate **360**, which in turn comprises a non-metal surface, e.g., a dielectric and/or a semiconductor surface. A first portion **370** of a thin film comprising TiN is formed on the substrate **360**, and a second portion **380** of the thin film is formed on the first portion **370**. The first and second portions **370**, **380** are formed by an atomic layer deposition method illustrated in FIG.

3A, in which the substrate 360 is exposed to first and second cycles with different corresponding precursor exposure pressures. Because the first portion 370 may be grown in a layer-by-layer growth mode in the initial stage, e.g., a nucleation stage as discussed above, at least the first portion 370 or both the first and second portions 370, 380 may be substantially free of adjacent crystals having different orientations characterized by columnar growth of V-shaped grains and the relatively high (e.g., 10% of the thickness) surface roughness. The resulting TiN layer has superior properties including one or more of relatively high conformality or step coverage, lower surface roughness, smaller average grain size, higher electrical conductivity and/or barrier characteristics relative to a comparable thin film layer formed at a single pressure during nucleation and bulk deposition stages.

[0054] As described herein and throughout the specification, it will be appreciated that the semiconductor substrate on which the TiN thin films according to embodiments can be implemented in a variety of substrates, including, but not limited to, a doped semiconductor substrate, which can be formed of an elemental Group IV material (e.g., Si, Ge, C or Sn) or an alloy formed of Group IV materials (e.g., SiGe, SiGeC, SiC, SiSn, SiSnC, GeSn, etc.); Group III-V compound semiconductor materials (e.g., GaAs, GaN, InAs, etc.) or an alloy formed of Group III-V materials; Group II-VI semiconductor materials (CdSe, CdS, ZnSe, etc.) or an alloy formed of Group II-VI materials.

[0055] According to certain embodiments, the substrate can also be implemented as a semiconductor on insulator, such as silicon on insulator (SOI), substrate. An SOI substrate typically includes a silicon-insulator-silicon structure in which the various structures described above are isolated from a support substrate using an insulator layer such as a buried SiO₂ layer. In addition, it will be appreciated that the various structures described herein can be at least partially formed in an epitaxial layer formed at or near a surface region.

[0056] Furthermore, the substrate can include a variety of structures formed thereon, e.g., diffusion regions, isolation regions, electrodes, vias and lines to name a few, on which any structure comprising the TiN layer according to embodiments may be formed, including topological features such as vias, cavities, holes or trenches having one or more semiconductor or dielectric surfaces. Thus, the non-metal surface on which the TiN layer according to embodiments is formed can include a semiconductor surface, e.g., a doped or undoped Si surface, and/or a dielectric surface, e.g., an interlayer dielectric (ILD) surface, a mask or a hard mask surface or a gate dielectric surface, to name a few, which can include an inorganic insulator, an oxide, a nitride, a high K dielectric, a low K dielectric, or carbon, to name a few dielectric materials.

[0057] As described herein and throughout the specification, a reactor chamber refers to any reaction chamber including a single wafer processing reaction chamber or a batch wafer processing reaction chamber that is suitably configured for thermal atomic layer deposition (ALD). In a thermal ALD reactor, the substrate may be placed on a suitable substrate holder, such as a susceptor or a carrier boat. The substrate may be directly heated by conduction through a heated susceptor, or indirectly heated by radiation from a radiation source such as a lamp or by convection through a heated chamber wall.

[0058] Generally in an ALD process, reactants or precursors, e.g., oxidizing and reducing reactants, are alternately introduced into a reaction chamber having disposed therein a substrate. The introduction of one or more reactants or precursors may be in turn be alternated with a purge and/or a pump out process for removing excess reactants or precursors from the reaction chamber. The reactants may be introduced into the reaction chamber under a condition over a suitable period of time such that the surface of the substrate becomes at least partly saturated with the precursors or reactants and/or a reaction product of the reactants. Excess or residual precursors or reactants may then be removed from the substrate, such as by being purged and/or pumped out of the reaction chamber. A pump out process may be performed by a suitable vacuum pumping process and a purge step may be performed by introducing a non-reactive or an inert gas, e.g., nitrogen or a noble

gas, into the reaction chamber. In the context of layers formed by thermal ALD in examples hereinbelow, there are generally two categories of precursors or reactants, namely nitrogen (N) precursors and titanium (Ti) precursors.

[0059] In the following, with reference to FIG. 4, example implementations of the method **300** (FIG. 3A) of forming by ALD, e.g., thermal ALD, a thin film comprising TiN having at least two regions formed by exposing a substrate to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments.

Atomic Layer Deposition of TiN by Exposing Substrate to a Plurality of Cycles with Different Corresponding Precursor Exposure Pressures

[0060] Referring back to FIG. 3A, after providing **310** the substrate (substrate **360** in FIG. 3B) comprising a non-metal surface in the reaction chamber, the method **300** proceeds to forming **320** by atomic layer deposition (ALD), e.g., thermal ALD, on the non-metal surface a first portion of the thin film by exposing the semiconductor substrate to one or more first ALD cycles, followed by a second portion of the thin film by exposing the semiconductor substrate to one or more second ALD cycles. In the following, exposure pressures applied during the first and second ALD cycles are described diagrammatically.

[0061] FIG. 4 diagrammatically illustrates pressure traces corresponding to exposures of the substrate to Ti and N precursors during a first cycle **400A** or stage, e.g., a nucleation stage, for forming the first portion **370** (FIG. 3B) of the thin film and the second cycle **400B** or stage, e.g., a bulk growth stage, for forming the second portion **380** (FIG. 3B) of the thin film, according to various embodiments. Referring to FIG. 4, a first portion of the thin film is formed by exposing the semiconductor substrate to one or more first ALD cycles **400A** each comprising one or more exposures **404** or exposure pulses to a partial pressure of a first Ti precursor and one or more exposures **408** or exposure pulses to a partial pressure of a first N precursor. A second portion of the thin film is formed by exposing the semiconductor substrate to one or more second ALD cycles **400B** each comprising one or more exposures **412** or exposure pulses to a partial pressure of a second Ti precursor and one or more exposures **416** or exposure pulses to a partial pressure of second N precursor.

[0062] As schematically depicted, each of the exposures **404** to a first Ti precursor, **408** to a first N precursor, **412** to a second Ti precursor and **416** to a second N precursor may have different partial pressure regimes, including a corresponding partial pressure rise regime **404A**, **408A**, **412A** and **416A**, a main exposure regime **404B**, **408B**, **412B** and **416B**, and a partial pressure fall regime **404C**, **408C**, **412C** and **416C**. Each of the partial pressure rise regimes **404A**, **408A**, **412A** and **416A** may correspond, for example, the respective precursor being introduced into the reaction chamber. Each of the main exposure regimes **404B**, **408B**, **412B** and **416B** may correspond to a period during which the amount of the respective precursor in the reaction chamber is relatively constant. The relative constant amount of the respective precursor may be maintained, e.g., using a pressure transducer or a throttle valve. Each of the partial pressure fall regime **404C**, **408C**, **412C** and **416C** may correspond, for example, to a regime when the respective precursor is being purged or pumped out of the reaction chamber.

[0063] Still referring to FIG. 4, it will be appreciated that, in some implementations, the precursors may be pumped out and/or purged out after each exposure. In some implementations where the precursor may be pumped out without being purged, the reaction chamber pressure may be substantially represented by the partial pressures of the respective precursors, and the pressures traces of the exposures **404**, **408**, **412** and **416** may substantially represent the reaction chamber pressures or the precursor partial pressures during the respective exposures. In some implementations where the precursor is purged out with an inert gas without being pumped out, the reaction chamber pressure may be represented by total reaction chamber pressures **404P**, **408P**, **412P** and **416P** corresponding to the exposures **404**, **408**, **412** and **416**, where the total reaction chamber pressures result from a mixture of the respective precursors and an inert gas.

[0064] In practice, a combination of pumping and purging may be used for higher throughput and improved film quality. In these implementations, the substrate may be subjected to partial pressures of the first Ti precursor, the first N precursor, the second Ti precursor and the second N precursor, while measuring total pressures **404P**, **408P**, **412P** and **416P** including during purging and pumping. In some embodiments, the total chamber pressure may be kept relatively constant throughout a given precursor exposure or exposure pulse while pumping power is adjusted using a pressure transducer and replacing the removed precursor with an inert gas. In these implementations, one or more first ALD **400A** cycles for forming the first portion (**370** in FIG. **3B**) may each comprise one or more exposures **404** to a partial pressure of a first Ti precursor (while the measured parameter may be a total reaction chamber pressure **404P**), and one or more exposures **408** to a partial pressure of a first N precursor (while measured parameter may be a total reaction chamber **408P**). Similarly, one or more second ALD cycles **400B** for forming the second portion (**380** in FIG. **3B**) may each comprise one or more exposures **412** to a partial pressure of a second Ti precursor (while the measured parameter may be a total pressure **412P**) and one or more exposures **416** to a partial pressure of a second N precursor (while the measured parameter may be a total pressure **416P**).

[0065] According to various embodiments, during an exposure to a precursor, the measured total reaction chamber pressure may be proportional to the partial pressure of the precursor. Thus, the total pressures **412P** and **416P** that are higher relative total pressures **404P** and **408P**, respectively, correspond to higher partial pressures of the second Ti precursor and the second N precursor relative to partial pressures of the first Ti precursor and the first N precursor, respectively. However, embodiments are not so limited and in other embodiments, the total pressures **412P** and **416P** that are higher relative total pressures **404P** and **408P**, respectively, may correspond to the same or lower partial pressures of the second Ti precursor and the second N precursor relative to partial pressures of the first Ti precursor and the first N precursor, respectively.

[0066] Referring back to the illustrated method **300** in FIG. **3A**, one or both of exposure pressures of the second Ti precursor and the second N precursor during one or more second ALD cycles of the latter, e.g., a bulk deposition phase are higher relative to corresponding one or both of exposure pressures of the first Ti precursor and the first N precursor during one or more first ALD cycles of the initial, e.g., a nucleation phase. In some embodiments, the exposure pressures may be partial pressures of the precursors or total pressures of the reaction chambers. Thus, in various embodiments, in reference to FIG. **4**, one or both of the exposure **412** to the second Ti precursor and the exposure **416** to the second N precursor may be at higher partial pressures and/or a higher total reaction chamber pressures relative to the corresponding one or both of the exposure **404** to a the first Ti precursor and the exposure **408** to the first N precursor, respectively.

[0067] Still referring to FIG. **4**, in various embodiments, the corresponding partial or total pressures between corresponding exposures to Ti and N precursors during the first and second cycles **400A** and **400B** may be corresponding partial or total pressures during any one of partial pressure rise regimes **404A**, **408A**, **412A** and **416A**, main exposure regimes **404B**, **408B**, **412B** and **416B**, and partial pressure fall regimes **404C**, **408C**, **412C** and **416C**. For example, the exposures **412**, **416** to one or both of the second Ti precursor and the second N precursor during the main exposure regimes **412B** and **416B**, respectively, during the second ALD cycle **400B**, may be at higher total or partial pressures relative to the exposures **404**, **408** to one or both of the first Ti precursor and the first N precursor during the main exposure regimes **404B** and **408B**, respectively, during the first ALD cycle **400A**. In various other embodiments, the corresponding partial or total pressures between corresponding exposures to Ti and N precursors during the first and second cycles **400A** and **400B** may be corresponding average, mean or peak partial or total pressures during the exposures **404**, **408**, **412** and **416**.

[0068] Still referring to FIG. **4**, in the illustrated embodiment, the total and/or partial pressures during the exposure **404** to the first Ti precursor and the exposure **408** to the first N precursor are

different, and the total and/or partial pressures during the exposure **412** to the second Ti precursor and the exposure **416** to the second N precursor are different. However, embodiments are not so limited, and in some embodiments, the total and/or partial pressures during the exposure **404** to the first Ti precursor and the exposure **408** to the first N precursor may be kept constant, and/or the total and/or partial pressures during the exposure **412** to the second Ti precursor and the exposure **416** to the second N precursor may be kept constant.

[0069] Still referring to FIG. 4, each of the total pressures during the exposure **404** to the first Ti precursor and the exposure **408** to the first N precursor, which may be the same or different, may be 0.01-0.2 torr, 0.2-0.4 torr, 0.4-0.6 torr, 0.6-0.8 torr, 0.8-1.0 torr, 1.0-1.5 torr, 1.5-2.0 torr, 2.0-2.5 torr, 2.5-3.0 torr, or a pressure in range defined by any of these values. Each of the total pressures during the exposure **412** to the second Ti precursor and the exposure **416** to the second N precursor, which may be the same or different, may be 3.0-4.0 torr, 4.0-5.0 torr, 5.0-6.0 torr, 6.0-7.0 torr, 7.0-8.0 torr, 8.0-9.0 torr, 9.0-10.0 torr, 10.0-11.0 torr, 11.0-12.0 torr, or a pressure in range defined by any of these values. A ratio of the total pressure (measured in torr) of the reaction chamber during the exposure **412** to the second Ti precursor and the exposure **404** to the first Ti precursor may be 2-5, 5-10, 10-20, 20-50, 50-100, or in a range defined by any of these values. Similarly a ratio of the total pressure of the reaction chamber during the exposure **416** to the second N precursor and the exposure **408** to the first N precursor may be 2-5, 5-10, 10-20, 20-50, 50-100, or in a range defined by any of these values. In each of the exposures **404**, **408**, **412** and **416**, the respective Ti or N precursor can make up 1-2%, 2-5%, 5-10%, 10-20%, 20-50%, 50-100% of the total amount of gas molecules in the reaction chamber, or a percentage in a range defined by any of these values.

[0070] Still referring to FIG. 4, according to various embodiments, the total pressure or the partial pressure during the exposure **404** to the first Ti precursor and the exposure **408** to the first N precursor, in conjunction with the flow rates of the respective precursors and an inert gas and the pumping power of the reaction chamber, are controlled such that the deposition rate during the first cycle **400A** or stage is between 0.10-0.20 Å/cycle, 0.20-0.30 Å/cycle, 0.30-0.40 Å/cycle, 0.40-0.50 Å/cycle, 0.50-0.60 Å/cycle or a value in a range defined by any of these values, per cycle including an exposure **404** to the first Ti precursor and an exposure **408** to the first N precursor. The total pressure or the partial pressure during the exposure **412** to the second Ti precursor and the exposure **416** to the second N precursor, in conjunction with the flow rates of the respective precursors and an inert gas and the pumping power of the reaction chamber are controlled such that the deposition rate during the second cycle **400B** or stage is 0.20-0.30 Å/cycle, 0.30-0.40 Å/cycle, 0.40-0.50 Å/cycle, 0.50-0.60 Å/cycle, 0.60-0.70 Å/cycle, 0.60-0.70 Å/cycle, 0.70-0.80 Å/cycle or a value in a range defined by any of these values, per cycle including an exposure **404** to the first Ti precursor and an exposure **408** to the first N precursor. A ratio of the deposition rate per cycle during the second cycle **400B** to the deposition rate per cycle during the first cycle **400A** may be 1-1.5, 1.5-2.0, 2.5-3.0, or a ratio in a range defined by any of these values.

[0071] The inventors have found that various technical advantages of TiN thin films disclosed herein may be realized when each of forming **320** (FIG. 3A) the first portion **370** (FIG. 3B) and forming **330** (FIG. 3A) the second portion **380** (FIG. 3B) of the thin film comprising TiN comprises exposing a semiconductor substrate to 1-25 cycles, 26-50 cycles, 50-100 cycles, 100-200 cycles, 200-300 cycles, 300-400 cycles, 400-500 cycles, 500-600 cycles, or a value in a range defined by any of these values, of the first cycles **400A** (FIG. 4) and second cycles **400B** (FIG. 4), respectively. According to various embodiments, a ratio of the number of second cycles to the number of first cycles can be greater than 1, 2, 5 or 10 or a ratio in a range defined by any of these values, or smaller than 1, 0.5, 0.1 or 0.1 or a ratio in a range defined by any of these values. The overall thickness of the thin film comprising TiN including the first portion **370** (FIG. 3B) and the second portion **380** (FIG. 3B) can have a combined stack thickness that does not exceed about 25 nm, 20 nm, 15 nm, 10 nm, 7 nm, 4 nm, 2 nm, or having a value in a range defined by any of these values. A thickness ratio between the first portion **370** (FIG. 3B) and the second portion **380** (FIG.

3B) can be about 1:20-1:10, 1:10-1:5, 1:5-1:2, 1:2-1:1, 1:1-2:1, 2:1-5:1, 5:1-10:1, 10:1-20:1, or a ratio in a range defined by any of these values. It will be appreciated that in some embodiments, the first portion **370** (FIG. 3B) may be relatively thinner, e.g., when higher conformality may be more important compared to lower film roughness, while in other embodiments, the second portion **380** (FIG. 3B) may be relatively thinner, e.g., when lower film roughness may be more important compared to higher conformality.

[0072] Still referring to FIG. 4, each of the exposure **404** of the substrate to the first Ti precursor and the exposure **412** of the substrate to the second Ti precursor is such that the surface of the substrate is substantially or partly saturated with the first Ti precursor or the second Ti precursor, respectively. After each of the exposure **404** of the substrate to the first Ti precursor and the exposure **412** of the substrate to the second Ti precursor, excess or residual first and/or second Ti precursors or their reaction products that do not remain adsorbed or chemisorbed on the surface of the substrate may be pumped and/or purged out.

[0073] Similarly, each of the exposure **408** of the substrate to the first N precursor and the exposure **416** of the substrate to the second N precursor is such that the substrate is substantially or partly saturated with the first N precursor or the second N precursor, respectively. After each of the exposure **408** of the substrate to the first N precursor and the exposure **416** of the substrate to the second N precursor, excess or residual first and/or second N precursors or their reaction products that do not remain adsorbed or chemisorbed on the surface of the substrate may be pumped and/or purged out. Subjecting the substrate to one or more exposures to the first Ti precursor and one or more exposures to the first N precursor may form about a monolayer or less per cycle of TiN. Similarly, subjecting the substrate to one or more exposures to the second Ti precursor and one or more exposures to the second N precursor may form about a monolayer or less per cycle of TiN.

[0074] In some embodiments, the exposure **404** to the first Ti precursor, the exposure **408** to the first N precursor, the exposure **412** to the second Ti precursor and/or the exposure **416** to the second N precursor may be performed a plurality of times in sequence prior to introduction of the other precursor. For example, advantageously, under some circumstances, exposing the substrate to a Ti precursor and/or a N precursor more than once may result in a higher level of surface saturation e.g., when substantial steric hindrance effect exists.

[0075] Still referring to FIG. 4, it will be appreciated that the relative order of exposures to a first Ti precursor and to a first N precursor may be selected depending on competing circumstances. In some implementations, the first Ti precursor may advantageously be the first precursor the substrate surface is exposed to. For example, one or more direct exposures of a Si surface to the first Ti precursor may lead to formation of one or more monolayers of TiSi and prevent formation of SiN, which may in turn be advantageous for lowering contact resistance between the underlying Si and the TiN layer formed thereover. However, in some other implementations, the first N precursor may advantageously be the first precursor the substrate is exposed to. For example, by directly exposing a Si surface to the first N precursor, one or more monolayers of SiN may be intentionally formed, which may be advantageous for improving barrier characteristics of the stack.

[0076] It will be appreciated that in various embodiments, the frequency and repetition of the exposures of the substrate to the first Ti reactant and/or the first N precursor in each of the first cycles **408A**, and to the second Ti reactant and/or the second N precursor in each of the second cycles **408B** may be varied to obtain a desired thickness and stoichiometry, based on various considerations including susceptibility to steric hindrance effects of the precursors.

[0077] According to various embodiments, non-limiting examples of the first and second Ti precursors, which may be the same or different for forming first and second portions of a TiN layer according to embodiments, include titanium tetrachloride (TiCl₄), tetrakis(dimethylamino)titanium (TDMAT) or tetrakis(diethylamino)titanium (TDEAT). Having same precursors for the first and second portions of the TiN may be advantageous for, e.g., being lower in cost and/or easier for process design. However, having different precursors for the first

and second portions of the TiN may be advantageous for, e.g., different deposition characteristics or film quality.

[0078] According to various embodiments, non-limiting examples of the first and second N precursors, which may be the same or different for forming first and second portions of a TiN layer according to embodiments, include ammonia (NH₃), hydrazine (N₂H₄) or monomethylhydrazine (CH₃(NH)NH₂, “MMH”). Having same precursors for the first and second portions of the TiN may be advantageous for, e.g., being lower in cost and/or easier for process design. However, having different precursors for the first and second portions of the TiN may be advantageous for, e.g., different deposition characteristics or film quality.

[0079] According to various embodiments, non-limiting examples of the inert gas for purging may include nitrogen N₂ or a noble gas such as Ar or He.

[0080] Various technical advantages and benefits described herein can be realized when one or both of the first and second portions **370, 380** (FIG. 3B) of the thin film comprising TiN are formed at a substrate temperature of 350° C. to 800° C., 450° C. to 750° C., 500° C. to 700° C., 550° C. to 650° C., or in a range defined by any of these values, for instance about 600° C., according to embodiments. Keeping the temperature the same during growth of the first and second portions **370, 380** may be advantageous for throughput an ease of process control, as temperature adjustments during process may take long time.

[0081] In various embodiments, the exposure times or pulse times of each of the first and second Ti precursors and first and second N precursors can be in the range of about 0.1-1 sec., 1-10 sec., 10-30 sec., 30-60 sec., or a duration in a range defined by any of these values.

[0082] Advantageously, when the TiN layer is formed using an atomic layer deposition method in which a substrate is exposed to a plurality of cycles with different corresponding precursor exposure pressures according to various embodiments, one or both of the surface roughness and electrical resistivity can be substantially reduced to conventional TiN films including TiN films formed using other ALD processes with a single pressure setpoint. As deposited, the thin film comprising TiN formed at according the methods described herein and having the above-indicated thicknesses and thickness ratios between the first and second portions **370, 380** (FIG. 3B) can have a root mean square (RMS) surface roughness of 3%, 4%, 5%, 6%, 7%, 8%, and 9%, or a value in a range defined by any of these values, on the basis of an average thickness of the thin film.

Alternatively, as-deposited, the thin film comprising TiN having the above-indicated thicknesses and thickness ratios between the first and second portions **370, 380** (FIG. 3B) can have an RMS surface roughness value that is less than 2.5 nm, 2 nm, 1.5 nm, 1.0 nm, 0.5 nm, or a value in a range defined by any of these values.

[0083] As deposited, the thin film comprising TiN formed according to the methods described herein and having the above-indicated thickness and thickness ratios between the first and second portions **370, 380** (FIG. 3B) can have an electrical resistivity of <70 μΩ-cm, 70-100 μΩ-cm, 100-130 μΩ-cm, 130-160 μΩ-cm, 160-190 μΩ-cm, 190-220 μΩ-cm, 220-250 μΩ-cm, 250-280 μΩ-cm, 280-310 μΩ-cm, or greater than 310 μΩ-cm, or a value in a range defined by any of these values, for instance less than about 200 μΩ-cm.

[0084] In addition to reduced surface roughness and electrical resistivity, the thin film comprising TiN formed according to the methods disclosed herein has high conformality when deposited in high aspect ratio structures. One measure of conformality in the context of high aspect ratio structures is referred to herein as step coverage. A high aspect ratio structure may be, e.g., a via, a hole, a trench, a cavity or a similar structure. By way of an illustrative example, FIG. 5 schematically illustrates a semiconductor structure **500** having an example high aspect ratio structure **516** formed therein, to illustrate some example metrics of defining and/or measuring conformality of thin films formed on high aspect ratio structures. The illustrated high aspect ratio structure **516** is lined with a TiN layer **512** having different thicknesses at different portions thereof. As described herein, a high aspect ratio structure has an aspect ratio, e.g., a ratio defined as a depth

or height (H) divided by a width (W) at the opening region of the high aspect ratio structure **516**, that exceeds **1**. In the illustrated example, the high aspect ratio structure **516** is a via formed through a dielectric layer **508**, e.g., an intermetal dielectric (ILD) layer, formed on a semiconductor substrate **504**, such that a bottom surface of the high aspect ratio structure **516** exposes the underlying semiconductor **504**. The TiN layer **512** can coat different surfaces of the high aspect ratio structure **516** with different thicknesses. As described herein, one metric for defining or measuring the conformality of a thin film formed in a high aspect ratio is referred to as step coverage. A step coverage may be defined as a ratio between a thickness of a thin film at a lower or bottom region of a high aspect ratio structure and a thickness of the thin film at an upper or top region of the high aspect ratio structure. The upper or top region may be a region of the high aspect ratio structure at a relatively small depth at, e.g., 0-10% or 0-25% of the H measured from the top of the opening. The lower or bottom region may be a region of the high aspect ratio structure at a relatively high depth at, e.g., 90-100% or 75-100% of the H measured from the top of the opening. In some high aspect ratio structures, a step coverage may be defined or measured by a ratio of thicknesses of the thin film **512A** formed at a bottom surface to the thin film **512C** formed at upper or top sidewall surfaces of the high aspect ratio structure. However, it will be appreciated that some high aspect ratio structures may not have a well-defined bottom surface or a bottom surface having small radius of curvature. In these structures, a step coverage may be more consistently defined or measured by a ratio of thicknesses of the thin film **512B** formed at a lower or bottom sidewall surface to the thin film **512C** formed at an upper or top sidewall surfaces of the high aspect ratio structure.

[0085] As described above, the thin film comprising TiN formed according to the methods disclosed herein results in a reduced surface roughness and electrical resistivity, while also providing high conformality in high aspect ratio structures. According to various embodiments, high aspect ratio structures having an aspect ratio exceeding 1, 2, 5, 10, 20, 50, 100, 200 or a value in a range defined by any of these values may be conformally coated with TiN films according to embodiments with a step coverage as defined herein that exceeds 70%, 80%, 90%, 95%, or has a value in a range defined by any of these values.

Physical Characterization of TiN Formed by Exposing Substrate to a Plurality of Cycles with Different Corresponding Precursor Exposure Pressures

[0086] FIG. **6** is a graph illustrating experimentally measured root mean square (RMS) surface roughness trend **604** and step coverage trend **608** as functions of the number of first cycles of exposures to Ti and N precursors at a relatively low chamber pressure of 0.5 torr, out of a total of 600 combined first cycles (e.g., nucleation stage) and second cycles (e.g., bulk deposition stage). The second cycles of exposures to Ti and N precursors were at a relatively high chamber pressure of 5 torr. Each experimental data point in FIG. **6** was taken from a TiN film grown on a native SiO₂-coated Si substrate for surface roughness measurements, and a TiN film grown in a via formed in SiO₂ and having about 40:1 aspect ratio. The measured deposition rates of the first and second cycles were 0.28 Å/cycle and 0.38 Å/cycle, respectively. The experimental data were measured on four different TiN films grown with 0 first cycles (0 Å)/600 second cycles (228 Å), 50 first cycles (14 Å)/550 second cycles (209 Å), 200 first cycles (56 Å)/400 second cycles (152 Å), and 600 first cycles (168 Å)/0 second cycles (0 Å). The four TiN films had total thicknesses of about 228 Å, 223 Å, 208 Å and 168 Å, respectively. As described above, the measured surface roughness values of the TiN films decrease with increasing relative number of first cycles including exposures at the relatively lower pressure. Without being bound to any theory, this may be because lower growth rate tends to allow for more surface diffusion which tends to reduce the surface roughness and promote layer-by-layer growth. The measured surface roughness values for thin films grown with 0 first cycles/600 second cycles, 50 first cycles/550 second cycles and 200 first cycles/400 second cycles were about 21 Å, 17.5 Å and 12.5 Å, respectively, corresponding to about 9%, 8% and 6% on the basis of the total thickness of the respective TiN films. In addition, as

discussed above, the measured step coverage values of the TiN films were higher for the thin film grown with 0 first cycles/600 second cycles relative to the film grown with 600 first cycles/0 second cycles. Without being bound to any theory, this may be because higher pressure tends to allow for more precursors to reach the bottom of the high aspect ratio, which tends to improve step coverage. However, surprisingly, the inventors have found that, up to about 50 first cycles (8% of total number of cycles), the increasing number of first cycles actually improved the step coverage. Thus, according to some embodiments, forming the first portion of the TiN film comprises alternately exposing the semiconductor substrate to 1 to 50 cycles each comprising an exposure to a first Ti precursor and an exposure to a first N precursor at a relatively low exposure pressure of less than about 3 torr.

[0087] FIGS. 7A-9 illustrate further experimental comparisons between a TiN film grown by exposing a substrate to cycles with the same precursor exposure pressures and a TiN film grown by exposing a substrate to a plurality of cycles with different corresponding precursor exposure pressures, according to embodiments. FIG. 7A is a cross-sectional transmission electron micrograph of high aspect ratio vias lined with a TiN layer formed by an atomic layer deposition method in which a substrate is exposed to ALD cycles at the same precursor exposure pressure corresponding to second cycles. FIGS. 7B and 7C are transmission electron micrographs (TEMs) of a TiN film grown using only second cycles of exposures to Ti and N precursors at a relatively high chamber pressure of 5 torr. The TEMs are images of a via formed in SiO₂ having about 40:1 aspect ratio, taken at upper (FIG. 7B) and lower (FIG. 7C) regions of the via. In comparison, FIGS. 8A and 8B are transmission electron micrographs (TEMs) of a TiN film grown using a combination of first and second cycles of exposures to Ti and N precursors at a relatively low (0.5 torr) and high (5 torr) chamber pressures, according to embodiments. The TEMs are images of a via formed in SiO₂ having about 40:1 aspect ratio, taken at upper (FIG. 8A) and lower (FIG. 8B) regions of the via. FIG. 9 is a graph illustrating an experimental statistical comparison between measured step coverages 904 measured from TEM micrographs shown in FIGS. 7A-7C and measured step coverages 908 measured from TEM micrographs shown in FIGS. 8A-8B. The data points in FIG. 9 represents a ratio taken from different locations within a lower region of the via and different locations within an upper region of the via. While not readily apparent from the TEM images, the statistical comparison in FIG. 9 clearly illustrate a higher median step coverage of 93% for the TiN film deposited according to embodiments and 87% for the TiN film deposited using a single exposure pressure. In addition, the statistical spread of measured step coverage for the TiN film deposited according to embodiments is substantially smaller than that for the TiN film deposited using a single exposure pressure, indicating that the film roughness is significantly higher for the latter.

Atomic Layer Deposition of TiN Thin Films Having Increased (111) Crystalline Texture

[0088] TiN as disclosed herein have a face-centered cubic, NaCl-type lattice structure. As such, the surface atom density is highest when the surface has a (111) orientation, relative to (200) orientation. The fraction of the surface having a certain crystal orientation for TiN can be highly deposition condition-dependent.

[0089] For certain applications of TiN thin films, e.g., DRAM capacitor electrodes, to meet the increasing demands of aggressive scaling, the TiN films may need to be extremely thin, e.g., less than 30 nm, while simultaneously meeting a combination of stringent electrical and mechanical properties. For example, in addition to low electrical resistivity and high conformality as discussed above, there is a need for some TiN thin films to simultaneously meet a combination of stringent mechanical properties, e.g., relatively high density, hardness and modulus, in order to reduce the risk of integration failures.

[0090] A TiN film grown by atomic layer deposition can have grains that have different crystal orientations including (111) and (200) orientations at the surface, among other orientations. The inventors have discovered that TiN thin films that are textured to have certain crystalline texture

can have superior mechanical properties, in addition to having a desirably low electrical resistivity and conformality as described above. In particular, ultrathin TiN films having relatively high (111) crystalline texture can have a relatively high density, hardness and modulus. In addition, the increased (111) crystalline texture can reduce columnar growth, thereby providing superior diffusion barrier properties. Without being bound to any theory, these advantageous properties of the TiN thin films having relatively high (111) crystalline texture may be associated with one of the highest surface atom packing density in a direction parallel to the growth surface, as well as grain boundary arrangements that favor the superior physical properties. The inventors have further discovered that the texture of TiN thin films can be controlled by controlling certain conditions of the cyclical vapor deposition cycles, as described herein. In particular, the inventors have discovered that cyclical vapor deposition cycles in which the substrate is subjected to a relatively high flow rate of the N precursor can cause a TiN thin film to grow with a relatively high (111) crystalline texture, as described herein.

[0091] FIG. **11** is a flow chart schematically illustrating an atomic layer deposition method **1100** of forming a TiN layer having a relatively high (111) crystalline texture by exposing a substrate to a relatively high N precursor flow rate, according to embodiments. The method **1100** includes providing **1110** a semiconductor substrate. The method **1100** additionally includes exposing **1120** the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate. The method **1100** forms a TiN film having a relatively high (111) crystalline texture by exposing **1120** the semiconductor substrate to one or more cyclical vapor deposition cycles in which a ratio of the N precursor flow rate to the Ti precursor flow rate (N/Ti flow ratio) and/or the flow rate of the N precursor are substantially high compared to conventional methods. According to various embodiments, a ratio of the N precursor flow rate to the Ti precursor flow rate (N/Ti flow ratio) exceeds 3. According to various embodiments, the N precursor flow rate exceeds **500** sccm. It will be appreciated that the exposing **1120** and forming **1130** are not necessarily ordered as shown. Various implementations of the method **1100** are described herein.

[0092] Providing **1110** the semiconductor substrate can be similar to providing **310** a semiconductor substrate according to various examples described above with respect to FIGS. **3A** and **3B**, the details of which are not repeated herein for brevity. For example, the substrate can be patterned or non-patterned, and can have one or both of insulating and conducting surfaces as described above.

[0093] Exposing **1120** the semiconductor substrate can be similar to exposing the semiconductor substrate to one or more first or second thermal ALD cycles as described above with respect to respect to FIGS. **3A**, **3B** and **4**, the details of which is not repeated herein for brevity. For example, the exposure to a Ti precursor at a Ti precursor flow rate can be in accordance with the exposure **404** (FIG. **4**) to the first Ti precursor or the exposure **412** (FIG. **4**) to the second Ti precursor. Similarly, the exposure to a N precursor at a N precursor flow rate can be in accordance with the exposure **408** (FIG. **4**) to the first N precursor or the exposure **416** (FIG. **4**) to the second N precursor. The details of these processes are not repeated herein for brevity. According to various embodiments, exposing **1120** is such that a TiN thin film having a high (111) crystalline texture is formed, as described herein.

[0094] According to various embodiments, the Ti precursor and the N precursor can be any of the precursors described above. For example, the Ti precursor can be titanium tetrachloride (TiCl₄) and the N precursor can be ammonia (NH₃).

[0095] Various other process parameters including the substrate temperature, chamber pressure and exposure times, can be in accordance with various process parameters described above, the details of which are not repeated herein for brevity.

[0096] According to embodiments, exposing **1120** is such that a ratio of the N precursor flow rate to the Ti precursor flow rate (N/Ti flow ratio) exceeds 2. The ratio can exceed 2, 3, 5, 10, 20, 30,

40, 50, 60, 70, 80, 90, 100, or have a value in a range defined by any of these values.

[0097] According to embodiments, exposing **1120** is such that the N precursor flow rate exceeds 200 sccm. For example, the N precursor flow rate can exceed 200 sccm, 500 sccm, 1000 sccm, 2000 sccm, 3000 sccm, 4000 sccm, 5000 sccm, 6000 sccm, 7000 sccm, 8000 sccm, 9000 sccm, 10000 sccm, or have a value in a range defined by any of these values.

[0098] According to embodiments, exposing **1120** is such that the Ti precursor flow rate exceeds 100 sccm but does not exceed the flow rate of the N precursor, in accordance with the N/Ti flow ratio described above. For example, the Ti precursor flow rate can exceed 100 sccm, 200 sccm, 500 sccm, 1000 sccm, 2000 sccm, 3000 sccm, 4000 sccm, 5000 sccm, or have a value in a range defined by any of these values.

[0099] FIG. **12A** illustrates experimental X-ray diffraction spectra of the resulting TiN thin films having a relatively high (111) crystalline texture, according to embodiments. The X-ray diffraction spectra were taken on Examples 1-4 shown in TABLE 1 below. FIG. **12B** is a graph illustrating the experimental ratios of a peak height of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN obtained from the X-ray diffraction spectra of FIG. **12A**. The illustrated TiN thin films are shown as examples only, and the embodiments are not so limited. According to embodiments, the N precursor flow rate range and the Ti precursor flow rate range are such that the TiN thin film has a relatively high (111) crystalline texture relative to a comparable thin film (e.g., a TiN having a similar thickness) formed by exposing the semiconductor substrate to cyclical vapor deposition cycles each comprising an exposure to a different Ti precursor flow rate range outside of the Ti precursor flow rate range described above and an exposure to a different N precursor flow rate range outside of the N precursor flow rate range described above. The degree of the increased (111) texture can be measured using, e.g., X-ray diffraction. For example, the relative intensities of different X-ray peaks can provide at least a semi-quantitative indication of the degree of texturing. The TiN thin films formed according to embodiments advantageously have a relatively high (111) crystalline texture such that an X-ray spectrum of the thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to a peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation that exceeds 0.4. For example, this ratio can exceed 0.4, 0.6, 0.8, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0, or a value in a range defined by any of these values. Selected experimental precursor flow conditions and the corresponding measurements of the ratios are summarized in TABLE 1.

[0100] FIG. **13** is a graph of experimental thickness and resistivity measurements of TiN thin films having relatively high (111) crystalline texture, according to embodiments. For Examples 1-4 listed in TABLE 1, increasing the N/Ti flow ratio decreases the thickness for the same number of cycles. Unexpectedly, unlike conventional methods, increasing the N/Ti flow ratio, which in turn decreases the thickness, decreases a resistivity of the TiN thin film. Prior to the present disclosure, decreasing thickness had been associated with increasing resistivity, which had been correlated to relatively higher amounts of chlorine in relatively thin TiN thin films formed with TiCl₄. The illustrated TiN thin films are shown as examples only, and the embodiments are not so limited.

Advantageously, the resistivity of the TiN films formed according to embodiments is less than about 200 $\mu\Omega$ -cm or any value described above. Selected experimental precursor flow conditions and the corresponding measurements of the thickness and resistivity are summarized in TABLE 1.

TABLE-US-00001

TABLE 1	TiCl ₄	NH ₃	(111) X-ray Flow	Flow NH ₃ /TiCl ₄	Sheet Peak/(200) rate rate	Flow rate	Thickness	Resistivity	Resistance	X-Ray Peak	Example								
(sccm)	(sccm)	Ratio (Å)	($\mu\Omega$ cm)	(Ω /sq)	Ratio	1	140	500	3.6	268	159	59.3	0.43	2	140	1000	7.1	265	145
54.5	0.56	3	140	2000	14.3	261	136	52.2	0.75	4	140	4000	28.6	261	129	49.4	1.39		

[0101] FIG. **14** is a graph of experimental hardness and modulus measurements of TiN thin films having substantially the same thicknesses and a preferential (111) crystalline texture, according to embodiments. FIG. **15** is a graph of experimental hardness measurements of TiN thin films having

a relatively high (111) crystalline texture for two different thicknesses, according to embodiments. According to embodiments, increasing the N/Ti flow ratio, which in turn decreases the thickness, increases a Young's modulus and a hardness of the TiN thin film. The illustrated TiN thin films are shown as examples only, and the embodiments are not so limited. Advantageously, the hardness of the TiN films formed according to embodiments exceeds 6 GPa. For example, the hardness can exceed 6 GPa, 10 GPa, 14 GPa, 18 GPa, 22 GPa, 25 GPa, or have a value in a range defined by any of these values. Advantageously, the elastic or Young's modulus of the TiN films formed according to embodiments exceeds 150 GPa. For example, the Young's modulus can exceed 150 GPa, 170 GPa, 190 GPa, 210 GPa, 230 GPa, 250 GPa, 270 GPa, 300 GPa, or have a value in a range defined by any of these values.

[0102] According to the methods described herein, decreasing a deposition pressure increases the ratio of the peak height or the intensity of the X-ray diffraction peak corresponding to the (111) crystal orientation of TiN to the peak height or the intensity of the X-ray diffraction peak corresponding to the (200) crystal orientation of TiN. This is illustrated in FIG. 16, which illustrates experimental X-ray spectra of TiN thin films formed at different exposure pressures and having a preferential (111) crystalline texture, according to embodiments. The illustrated TiN thin films are shown as examples only, and the embodiments are not so limited. In particular, the X-ray diffraction profiles were taken for TiN films having the same thickness but formed at two different exposure pressures, namely 5 torr and 3 torr. The results show that lower pressure can lead to further increase in the (111) crystalline texture. The inventors have recognized, however, that under some circumstances, lower exposure pressure can lead to compromised step coverage. Accordingly, in these circumstances, according to embodiments, after forming **1130** (FIG. 11) the TiN thin film having a relatively high (111) crystalline texture, the method **1100** (FIG. 11) can further include forming on the TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor, wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more first cyclical vapor deposition cycles. Other exposure conditions corresponding to forming **1130** (FIG. 11) the TiN thin film and forming the second TiN thin film can be according to any of the exposure conditions described above with respect to FIGS. 3A, 3B and 4.

[0103] FIG. 17 illustrates chlorine concentration depth profiles of TiN thin films having increased (111) crystalline texture, according to embodiments. The depth profiles are obtained using secondary ion mass spectrometry. The measured thin films correspond to Examples 1-4 listed in TABLE 1. As described above, increasing the N/Ti flow ratio decreases the thickness for the same number of cycles. Unexpectedly, unlike conventional methods, increasing the N/Ti flow ratio, which in turn decreases the thickness, decreases a resistivity of the TiN thin film. FIG. 17 illustrates that this unexpected trend of decreasing resistivity with decreasing thickness correlates to decreasing chlorine concentration. For the range of N/Ti flow ratios covered by Examples 1-4, the amount of chlorine can decrease by more than 50%. For example, at about 10 nm, for Examples, 1-4, the chlorine concentration was measured to be $7.2 \times 10^{20}/\text{cm}^3$, $6.1 \times 10^{20}/\text{cm}^3$, $5.1 \times 10^{20}/\text{cm}^3$ and $3.5 \times 10^{20}/\text{cm}^3$, respectively. Prior to the present disclosure, decreasing thickness had been associated with increasing chlorine content, which had been correlated to relatively higher resistivity.

Atomic Layer Deposition of Dual Layer TiN Thin Films Having Increased (111) Crystalline Texture

[0104] As described above, high N/Ti flow ratios, high N precursor flow rate and/or low deposition pressure may increase the (111) texture of the resulting TiN thin film. As further described above, high (111) texture may provide various advantages in a similar manner to low pressure-deposited

(e.g., less than 5 torr) TiN thin films as described above. Such advantages include improvements in resistivity, hardness, modulus and roughness to name a few. However, the high N/Ti flow ratios may result in relatively low step coverage under some circumstances, in a similar manner as described above with respect to low pressure-deposited TiN thin films. To compensate, it may be desirable to have a dual or multiple layer process in which a second TiN thin film is grown on a first TiN thin film having a high (111) texture.

[0105] According to these embodiments, the method comprises forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate. The method additionally comprises forming on the TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate. The method is such that one or both of the TiN thin film and the second TiN thin film have high (111) crystalline texture such that X-ray spectra of the one or both of the TiN thin film and the second TiN thin film have a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0106] According to some other embodiments, a method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process comprises forming on a semiconductor substrate a first TiN thin film at a first pressure by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate. The first TiN thin film has a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4. The method additionally comprises forming on the first TiN thin film a second TiN thin film at a second pressure higher than the first pressure by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate. The second TiN thin film can be formed according to any first or second TiN thin film formed as part of a dual or multiple layer TiN thin film process described above, e.g., those described above with respect to FIGS. 3A, 3B and 4.

[0107] According to these embodiments, the resulting TiN thin film comprises a lower portion and an upper portion, where the lower and upper portions can have different crystalline textures. At least the lower portion of the thin film has a crystalline texture such that an X-ray diffraction profile of the thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to a peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation exceeds 0.4. The upper portion can have similar or lower degree of (111) texture, depending on the deposition condition and the desired final film characteristics, as described herein.

[0108] Advantageously, the second TiN thin film grown on the TiN thin film can benefit from a high (111) texture of the underlying first TiN thin film. Under these circumstances, the second TiN thin film can also have relatively high (111) texture despite that fact that, if grown on another surface that does not have the high (111) TiN texture of the underlying first TiN thin film, the second TiN thin film may not have a relatively high (111) texture. For example, if grown on a different surface, e.g., a material other than TiN or TiN that does not have a relatively high (111) texture, such as an insulating film such as SiO₂, Si or other materials, the second TiN thin film may not have a relatively high (111) texture. For example, if grown on another surface, the second TiN layer may have a crystalline texture such that an X-ray spectrum of the TiN thin film has a

ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that is substantially less than 0.4, e.g., 0.3, 0.2, 0.1 or a value in a range defined by any of these values. This is because the textured surface of the underlying TiN thin film can serve as a template for the second or subsequent TiN thin film. Because the second or subsequent TiN thin film is not necessarily formed under high (111) texturing condition, the second or subsequent TiN thin film can be formed under conditions that improve the overall performance of the combined TiN thin film. For example, the second or subsequent TiN thin film can be grown under conditions that increase the step coverage, e.g., at high pressure, as described above with respect to various two-step deposition processes.

[0109] For example, the second TiN thin film can be formed at a high pressure, e.g., greater than 3 or 5 torr, as described above, e.g., according to various two step processes described with respect to FIGS. 3A and 3B, with or without the high N precursor flow rate and/or N/Ti precursor flow ratio. For example, by texturing the first TiN thin film to have a relatively high (111) orientation, the initial film growth may proceed substantially in a (111) texture-favorable mode, which advantageously results in improved mechanical properties and electrical resistivity as described above. On the other hand, by subsequently exposing the substrate to the Ti and/or N precursors at relatively high pressures, e.g., greater than 3 or 5 torr, during deposition of the second portion of the thin film, the latter portion of the thin film, or the second TiN thin film, can advantageously grow with a degree of conformality or a step coverage that is higher relative to a the first TiN thin film having a relatively high (111) texture, or relative to a TiN thin film that is deposited by exposing the substrate to the Ti and/or N precursors at the relatively low pressures, e.g., less than 3 torr or less than 1 torr.

[0110] In addition, because the first portion of the TiN thin film has a relatively high (111) texture, the second portion of the thin film, which may continue to grow in a layer by layer fashion, using the first portion as a template. The second TiN thin film can have a lower or comparable (111) texture relative the underlying first TiN thin film.

[0111] According to various embodiments, the thickness ratio between first and second TiN thin films can be 10%, 25%, 50%, 25%, 90% or a value in a range defined by any of these values, depending on the final film characteristics desired. For example, where higher degree of (111) texture is desired, the first TiN thin film may be 50% or greater in thickness relative to the overall thickness, whereas where higher degree of conformality is desired, the second TiN thin film may be 50% or greater in thickness.

[0112] The resulting TiN thin film can have (111) texture that is any fraction of the (111) texture value of the first TiN thin film, e.g., 10%, 25%, 50%, 25%, 90% or a value in a range defined by any of these values. The resulting Young's modulus and/or hardness can also have a value defined by these values.

[0113] The resulting TiN thin film can have a conformality value that is any fraction of the conformality value of the second TiN thin film without the underlying first TiN thin film, e.g., 10%, 25%, 50%, 25%, 90% or a value in a range defined by any of these values thereof.

[0114] As a net result, when deposited on certain surfaces, e.g., a surface comprising a non-metallic surface, the thin film comprising the first and second portions deposited by depositing a relatively highly (111) textured first TiN thin film followed by a second TiN thin film according to methods disclosed herein advantageously has a combination of surface roughness and conformality that is superior relative to a thin film layer formed on the same surface using a single step. Alternatively, or in addition, in part owing to the improved smoothness and conformality, the thin film has a relatively low electrical resistivity compared to TiN layers formed by some existing methods.

Applications

[0115] The thin films comprising TiN formed using different exposure pressures according to various embodiments disclosed herein can be used in a variety of applications, particularly where

the substrate comprises a relatively high aspect ratio structure and/or a non-metal surface that can benefit from various advantageous characteristics of the TiN layer as disclosed herein. Example applications include deposition a via, a hole, a trench, a cavity or a similar structure having an aspect ratio, e.g., a ratio defined as a depth divided by a top width, that exceeds 1, 2, 5, 10, 20, 50, 100, 200 or a value in a range defined by any of these values.

[0116] By way of example, FIG. 10 schematically illustrates an application in the context of a diffusion barrier for a contact structure, e.g., a source or drain contact, formed on an active semiconductor substrate region that may be heavily doped. A portion of a semiconductor device 1000 is illustrated, which includes a substrate 1004 on which a dielectric layer 1008, e.g., an interlayer or intermetal dielectric (ILD) layer comprising a dielectric material such as an oxide or nitride is formed. In order to form contacts to various regions of the substrate 1004, including various doped regions, e.g., source and drain regions, a via or a trench may be formed through the dielectric layer 1008. The via or the trench may expose various non-metal surfaces, e.g., an exposed bottom surface comprising a substrate surface, e.g., a silicon substrate surface, as well as dielectric sidewalls of the vias. The bottom and side surfaces of the via can be conformally coated with a first portion (corresponding to the first portion 370 in FIG. 3B) of the TiN layer, followed by a second portion (corresponding to the second portion 380 in FIG. 3B) formed according to various embodiments described herein. A conformal first portion may first be formed directly on inner surfaces of the via, followed by formation of a conformal second TiN layer, according to various embodiments disclosed herein. Thereafter, the lined via may be filled with a metal, e.g., W, Al or Cu, to form a contact plug 1016. For example, the via may be filled with tungsten by CVD using, e.g., WF₆.

[0117] The barrier layer 1012 formed according to embodiments can be advantageous for various reasons. In particular, due to the conformal nature of the barrier layer 1012 formed by ALD, the propensity for a pinching off during the subsequent metal fill process may be substantially reduced. In addition, as described above, the barrier layer 1012 can provide effective hindrance of material transport thereacross, e.g., dopant (B, P) out-diffusion from the substrate 1004, as well as in-diffusion of reactants, etchants and metals (e.g., F, Cl, W or Cu) from the contact plug formation process. The barrier effect may be enhanced by reduced surface roughness and increased step coverage. Furthermore, as described above, a layer-by-layer growth mode may reduce the overall contact resistance of the barrier layer 1012. Furthermore, due to the reduced film roughness, a relatively thinner barrier layer 1012 may be formed while still accomplishing its desired barrier function, leading to further reduction in contact resistance.

[0118] Other applications of the TiN layers formed according to various embodiments disclosed herein include conductive structures formed in recessed substrates (e.g., buried electrodes or lines), electrodes (e.g., DRAM capacitor electrodes or gate electrodes), metallization barriers for higher metal levels (e.g., barriers in vias/trenches for Cu contacts/lines), high aspect ratio vertical rod electrodes or vias for three-dimensional memory and through-silicon vias (TSVs), to name a few.

EXAMPLE EMBODIMENTS I

[0119] 1. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0120] forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, [0121] wherein a ratio of the N precursor flow rate to the Ti precursor flow rate exceeds 3.

[0122] 2. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0123] forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate, [0124] wherein the N precursor flow rate exceeds 200

sccm.

[0125] 3. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0126] forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate within a Ti precursor flow rate range and an exposure to a N precursor at a N precursor flow rate within a N precursor flow rate range, [0127] wherein the N precursor flow range and the Ti precursor flow range are such that the TiN thin film has a preferential (111) crystalline texture relative to another TiN thin film formed by exposing the semiconductor substrate to cyclical vapor deposition cycles each comprising an exposure to the Ti precursor within a different Ti precursor flow rate range outside of the Ti precursor flow rate range and an exposure to the N precursor within a different N precursor flow rate range outside of the N precursor flow rate range.

[0128] 4. The method of any of the above Embodiments, wherein the ratio of the N precursor flow rate to the Ti precursor flow rate is 2-100.

[0129] 5. The method of any of the above Embodiments, wherein the N precursor flow rate is 200 sccm-10,000 sccm.

[0130] 6. The method of any of the above Embodiments, wherein the Ti precursor flow rate is 100 sccm-5000 sccm.

[0131] 7. The method of any of the above Embodiments, wherein the TiN thin film has a preferential (111) crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation exceeds 0.4.

[0132] 8. The method of any of the above Embodiments, wherein the TiN thin film has a hardness exceeding 6 GPa.

[0133] 9. The method of any of the above Embodiments, wherein the TiN thin film has a Young's modulus exceeding 150 GPa.

[0134] 10. The method of any of the above Embodiments, further comprising: [0135] forming on the TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor, [0136] wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles.

[0137] 11. The method of Embodiment 10, wherein the exposures to one or both of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles are at a reactor pressure of less than about 5 torr, and wherein the exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at a reactor pressure greater than about 5 torr.

[0138] 12. The method of Embodiments 10 or 11, wherein the exposure to each of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles is at a higher pressure relative to the corresponding exposure to each of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles.

[0139] 13. The method of any of Embodiments 10-12, wherein forming the TiN thin film comprises depositing at a first deposition rate less than 0.3 Å per one of the cyclical vapor deposition cycles comprising an exposure of the semiconductor substrate to the Ti precursor and an exposure to the N precursor, and wherein forming the second TiN thin film comprises depositing at a second deposition rate greater than 0.3 Å per one of the second cyclical vapor deposition cycles comprising an exposure of the semiconductor substrate to the second Ti precursor and an exposure to the second N precursor.

[0140] 14. The method of any of the above Embodiments, wherein a root mean square surface roughness of the thin film is less than about 8% of a thickness of the thin film.

[0141] 15. The method of any of Embodiments 10-14, wherein forming the TiN thin film and the second TiN thin film comprise depositing by thermal cyclical vapor deposition.

[0142] 16. The method of any of Embodiments 10-15, wherein forming the TiN thin film comprises directly exposing one or both of a semiconductor surface and an insulator surface to the one or more cyclical vapor deposition cycles.

[0143] 17. The method of any of Embodiments 10-16, wherein forming one or both of the TiN thin film and the second TiN thin film comprises growing in a layer-by-layer growth mode.

[0144] 18. The method of any of Embodiments 10-17, wherein forming the TiN thin film comprises alternately exposing the semiconductor substrate to 1 to 50 the cyclical vapor deposition cycles.

[0145] 19. The method of any of Embodiments 10-18, wherein forming the TiN thin film and the second TiN thin film comprises forming at a temperature between 400° C. and 600° C.

[0146] 20. The method of any of the above Embodiments, wherein the semiconductor substrate comprises a trench or a via comprising an inner surface comprising a non-metallic sidewall surface in a trench or a via having an aspect ratio exceeding 1, and wherein forming the thin film comprises conformally lining the inner surface, wherein a ratio of thicknesses of the thin film formed on lower 25% of a height of the trench or the via and upper 25% of the height of the trench or the via exceeds 0.9.

[0147] 21. The method of any of the above Embodiments, wherein an electrical resistivity of the thin film is less than about 200 $\parallel\Omega$ -cm.

[0148] 22. The method of any of Embodiments 10-21, wherein the Ti precursor is the same as the second Ti precursor and the N precursor is the same as the second N precursor.

[0149] 23. The method of any of Embodiments 1-9, the method comprising: [0150] providing the semiconductor substrate comprising a trench or a via having an aspect ratio exceeding 1; [0151] forming the thin film in the trench or the via by exposing the semiconductor substrate to the one or more cyclical vapor deposition to form the TiN thin film in the trench or the via; and [0152] further exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor and an exposure to a second N precursor to form a second TiN thin film on the TiN thin film, wherein exposures to one or both of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles are at different pressures relative to corresponding exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles.

[0153] 24. The method of Embodiment 23, wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles.

[0154] 25. The method of Embodiments 23 or 24, wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the Ti precursor and the N precursor during the one or more cyclical vapor deposition cycles by a factor of 5 or greater.

[0155] 26. The method of any of Embodiments 23-25, wherein forming the TiN thin film comprises alternately exposing the semiconductor substrate to 1 to 50 cyclical vapor deposition cycles.

[0156] 27. The method of any of Embodiments 23-26, wherein forming the TiN thin film comprises exposing the semiconductor substrate to a first number of the one or more cyclical vapor deposition cycles, and wherein forming the second portion comprises exposing the semiconductor substrate to a second number of second cyclical vapor deposition cycles greater than twice the first

number of the one or more cyclical vapor deposition cycles.

[0157] 28. The method of any of Embodiments 23-27, wherein a root mean square surface roughness of the thin film is less than about 8% of a thickness of the thin film.

[0158] 29. The method of any of Embodiments 23-28, wherein forming the TiN layer comprises forming by thermal cyclical vapor deposition at a temperature between 400° C. and 600 ° C.

[0159] 30. The method of any of Embodiments 23-29, wherein the trench or the via has an aspect ratio exceeding 5 and an inner surface comprising a non-metallic sidewall surface, and wherein forming the thin film comprises conformally lining the inner surface, wherein a ratio of thicknesses of the thin film formed on lower 25% of a height of the trench or the via and upper 25% of the height of the trench or the via exceeds 0.9.

[0160] 31. The method of any of Embodiments 23-29, wherein the Ti precursor is the same as the second Ti precursor and the N precursor is the same as the second N precursor.

[0161] 32. A semiconductor structure, comprising: [0162] a semiconductor substrate comprising a non-metallic sidewall surface in a trench or a via having an aspect ratio exceeding 5; and [0163] a thin film comprising TiN conformally lining the non-metallic sidewall surface, [0164] wherein the thin film has a preferential (111) crystalline texture such that an X-ray spectrum of the thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to (111) crystal orientation to a peak height or an intensity of an X-ray diffraction peak corresponding to (200) crystal orientation exceeds 0.4.

[0165] 33. The semiconductor structure of Embodiment 32, wherein the ratio exceeds 1.3.

[0166] 34. The semiconductor structure of Embodiments 32 and 33, wherein the thin film has a hardness exceeding 6 GPa.

[0167] 35. The semiconductor structure of any of Embodiments 32-34, wherein the thin film has a Young's modulus exceeding 150 GPa.

[0168] 36. The semiconductor structure of any of Embodiments 32-35, wherein a ratio of thicknesses of the thin film formed on lower 25% of a height of the trench or the via and upper 25% of the height of the trench or the via exceeds 0.9.

[0169] 37. The semiconductor structure of any of Embodiments 32-36, wherein the trench or the via has an aspect ratio exceeding 10.

[0170] 38. The semiconductor structure of any of Embodiments 32-37, wherein a root mean square surface roughness of the thin film formed on the non-metallic surface is less than about 8% on the basis of an average thickness of the thin film.

[0171] 39. The semiconductor structure of any of Embodiments 32-38, wherein the trench or the via has a dielectric sidewall.

[0172] 40. The semiconductor structure of any of Embodiments 32-39, wherein the trench or the via has a bottom surface exposing a semiconductor material of the semiconductor substrate.

[0173] 41. The semiconductor structure of any of Embodiments 32-40, wherein the thin film has an electrical resistivity less than about 200 $\mu\Omega$ -cm.

[0174] 42. The semiconductor structure of any of Embodiments 32-41, wherein the trench or the via lined with the thin film is filled with a metal comprising tungsten.

EXAMPLE EMBODIMENTS II

[0175] 1. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0176] forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, [0177] wherein a ratio of the N precursor flow rate to the Ti precursor flow rate (N/Ti flow ratio) exceeds 3, and [0178] wherein the method forms the TiN thin film having a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal

orientation of TiN that exceeds 0.4.

[0179] 2. The method of Embodiment 1, wherein the N/Ti flow ratio is 3-100.

[0180] 3. The method Embodiments 1 or 2, wherein the method is such that increasing the N/Ti flow ratio decreases a thickness of the TiN thin film.

[0181] 4. The method of any one of Embodiments 1-3, wherein increasing the N/Ti flow ratio decreases a resistivity of the TiN thin film.

[0182] 5. The method of any one of Embodiments 1-4, wherein the method is such that increasing the N/Ti flow ratio increases a Young's modulus of the TiN thin film.

[0183] 6. The method of Embodiment 5, wherein increasing the Young's modulus comprises increasing to a value exceeding 150 GPa.

[0184] 7. The method of any one of Embodiments 1-6, wherein the method is such that increasing the N/Ti flow ratio increases a hardness of the TiN thin film.

[0185] 8. The method of Embodiment 7, wherein increasing the hardness comprises increasing to a value exceeding 6 GPa.

[0186] 9. The method of any one of Embodiments 1-8, wherein increasing the N/Ti flow ratio decreases a chlorine content of the TiN thin film.

[0187] 10. The method of any one of Embodiments 1-9, wherein the method is such that decreasing a deposition pressure increases the ratio of the peak height or the intensity of the X-ray diffraction peak corresponding to the (111) crystal orientation of TiN to the peak height or the intensity of the X-ray diffraction peak corresponding to the (200) crystal orientation of TiN.

[0188] 11. The method of any one of Embodiments 1-10, wherein the N precursor flow rate is 500 sccm-10,000 sccm.

[0189] 12. The method of any one of Embodiments 1-11, wherein the Ti precursor flow rate is 100 sccm-5000 sccm.

[0190] 13. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0191] forming on a semiconductor substrate a TiN thin film by exposing the semiconductor substrate to one or more cyclical vapor deposition cycles each comprising an exposure to a Ti precursor at a Ti precursor flow rate and an exposure to a N precursor at a N precursor flow rate, [0192] wherein the N precursor flow rate exceeds 500 sccm, and [0193] wherein the method forms the TiN thin film having a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0194] 14. The method of Embodiment 13, a ratio of the N precursor flow rate to the Ti precursor flow rate (N/Ti flow ratio) is 3-100.

[0195] 15. The method of Embodiment 14, wherein the method is such that increasing the N/Ti flow ratio decreases a thickness of the TiN thin film.

[0196] 16. The method of Embodiment 15, wherein decreasing the thickness decreases a resistivity of the TiN thin film.

[0197] 17. The method of any one of Embodiments 15-16, wherein decreasing the thickness increases a Young's modulus of the TiN thin film.

[0198] 18. The method of Embodiment 17, wherein increasing the Young's modulus comprises increasing to a value exceeding 150 GPa.

[0199] 19. The method of any one of Embodiments 15-18, wherein decreasing the thickness increases a hardness of the TiN thin film.

[0200] 20. The method of Embodiment 19, wherein increasing the hardness comprises increasing to a value exceeding 6 GPa.

[0201] 21. The method of any one of Embodiments 15-20, wherein decreasing the thickness decreases a chlorine content of the TiN thin film.

[0202] 22. The method of any one of Embodiments 13-21, wherein decreasing a deposition

pressure increases the ratio of the peak height or the intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN.

[0203] 23. The method of any one of Embodiments 13-22, wherein the N precursor flow rate is 500 sccm-10,000 sccm.

[0204] 24. The method of any one of Embodiments 13-23, wherein the Ti precursor flow rate is 100 sccm-5000 sccm.

[0205] 25. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0206] forming on a semiconductor substrate a first TiN thin film by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate, [0207] forming on the first TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate, [0208] wherein the method is such that one or both or a combination of the first and second TiN thin films have a preferential (111) crystalline texture such that X-ray spectra of the one or both of the first and second TiN thin films have a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

[0209] 26. The method of Embodiment 25, wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles.

[0210] 27. The method of Embodiment 26, one or both of a first ratio of the first N precursor flow rate to the first Ti precursor flow rate (first N/Ti flow ratio) and a second ratio of the second N precursor flow rate to the second Ti precursor flow rate (second N/Ti flow rate) are 3-100.

[0211] 28. The method of Embodiment 27, wherein the method is such that increasing one or both of the first N/Ti flow ratio and the second N/Ti flow rate decreases corresponding thicknesses of one or both of the first and second TiN thin films.

[0212] 29. The method of Embodiments 28, wherein decreasing the corresponding thicknesses decreases corresponding resistivities of the one or both of the first and second TiN thin films.

[0213] 30. The method of Embodiments 28 or 29, wherein decreasing the corresponding thicknesses increases corresponding Young's moduli of the one or both of the first and second TiN thin films.

[0214] 31. The method of Embodiment 30, wherein increasing the Young's moduli comprises increasing to a value exceeding 150 GPa.

[0215] 32. The method of any one of Embodiments 28-31, wherein decreasing the corresponding thicknesses increases corresponding hardness values of the TiN thin film and the second TiN thin film.

[0216] 33. The method of Embodiment 32, wherein increasing the hardness values comprises increasing to a value exceeding 6 GPa.

[0217] 34. The method of any one of Embodiments 28-33, wherein decreasing the corresponding thicknesses decreases corresponding chlorine contents of the first and second TiN thin films.

[0218] 35. The method of any one of Embodiments 25-34, wherein the ratio of the peak height or the intensity of the X-ray diffraction peak corresponding to the (111) crystal orientation of TiN to the peak height or the intensity of the X-ray diffraction peak corresponding to the (200) crystal orientation of TiN is higher for the TiN thin film relative that for the second TiN thin film.

[0219] 36. The method of any one of Embodiments 25-35, wherein the exposures to one or both of

the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles are at a reactor pressure of less than about 5 torr, and wherein the exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at a reactor pressure greater than about 5 torr.

[0220] 37. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: [0221] forming on a semiconductor substrate a first TiN thin film at a first pressure by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate, [0222] wherein the first TiN thin film has a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4; and [0223] forming on the first TiN thin film a second TiN thin film at a second pressure higher than the first pressure by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate.

[0224] 38. The method of Embodiment 37, wherein the second pressure is greater than 5 torr.

[0225] 39. The method of Embodiment 37 or 38, wherein the second TiN thin films has a crystalline texture such that X-ray spectrum of the second TiN thin films has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that is lower than the corresponding ratio of the first TiN thin film.

[0226] 40. The method of any one of Embodiments 37-39, wherein at least a first ratio of the first N precursor flow rate to the first Ti precursor flow rate (first N/Ti flow ratio) is 3-100.

[0227] 41. The method of Embodiment 40, wherein a second ratio of the second N precursor flow rate to the second Ti precursor flow rate is (second N/Ti flow ratio) lower than the first N/Ti flow ratio.

[0228] 42. The method of Embodiment 41, wherein the method is such that increasing one or both of the first N/Ti flow ratio and the second N/Ti flow rate decreases corresponding thicknesses of one or both of the first and second TiN thin films.

[0229] 43. The method of Embodiment 42, wherein decreasing the corresponding thicknesses decreases corresponding resistivities of the one or both of the first and second TiN thin films.

[0230] 44. The method of Embodiment 42, wherein decreasing the corresponding thicknesses increases corresponding Young's moduli of the one or both of the first and second TiN thin films.

[0231] 45. The method of Embodiment 44, wherein increasing the Young's moduli comprises increasing to a value exceeding 150 GPa.

[0232] 46. The method of Embodiment 42, wherein decreasing the corresponding thicknesses increases corresponding hardness values of the one or both of the first and second TiN thin films.

[0233] 47. The method of Embodiment 46, wherein increasing the hardness values comprises increasing to a value exceeding 6 GPa.

[0234] 48. The method of Embodiment 42, wherein decreasing the corresponding thicknesses decreases corresponding chlorine contents of the first and second TiN thin films.

[0235] 49. The method of Embodiment 37, wherein the first pressure is less than 5 torr.

[0236] Although the present invention has been described herein with reference to the specific embodiments, these embodiments do not serve to limit the invention and are set forth for illustrative purposes. It will be apparent to those skilled in the art that modifications and improvements can be made without departing from the spirit and scope of the invention.

[0237] Such simple modifications and improvements of the various embodiments disclosed herein are within the scope of the disclosed technology, and the specific scope of the disclosed technology

will be additionally defined by the appended claims.

[0238] In the foregoing, it will be appreciated that any feature of any one of the embodiments can be combined or substituted with any other feature of any other one of the embodiments.

[0239] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise,” “comprising,” “include,” “including” and the like are to be construed in an inclusive sense, as opposed to an exclusive or exhaustive sense; that is to say, in the sense of “including, but not limited to.” The word “coupled”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Likewise, the word “connected”, as generally used herein, refers to two or more elements that may be either directly connected, or connected by way of one or more intermediate elements. Additionally, the words “herein,” “above,” “below,” and words of similar import, when used in this application, shall refer to this application as a whole and not to any particular portions of this application. Where the context permits, words in the above Detailed Description using the singular or plural number may also include the plural or singular number, respectively. The word “or” in reference to a list of two or more items, that word covers all of the following interpretations of the word: any of the items in the list, all of the items in the list, and any combination of the items in the list.

[0240] Moreover, conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” “for example,” “such as” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements and/or states. Thus, such conditional language is not generally intended to imply that features, elements and/or states are in any way required for one or more embodiments or whether these features, elements and/or states are included or are to be performed in any particular embodiment.

[0241] While certain embodiments have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel apparatus, methods, and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions and changes in the form of the methods and systems described herein may be made without departing from the spirit of the disclosure. For example, while features are presented in a given arrangement, alternative embodiments may perform similar functionalities with different components and/or sensor topologies, and some features may be deleted, moved, added, subdivided, combined, and/or modified. Each of these features may be implemented in a variety of different ways. Any suitable combination of the elements and acts of the various embodiments described above can be combined to provide further embodiments. The various features and processes described above may be implemented independently of one another, or may be combined in various ways. All possible combinations and subcombinations of features of this disclosure are intended to fall within the scope of this disclosure.

Claims

1. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: forming on a semiconductor substrate a first TiN thin film by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate; and forming on the first TiN thin film a second TiN thin film by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate, wherein the method is such that one or both of the first and second TiN thin films have a preferential (111) crystalline texture such that X-ray spectra of the one or both of the first and second TiN thin films

have a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4.

2. The method of claim 1, wherein exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at higher pressures relative to corresponding exposures to one or both of the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles.

3. The method of claim 2, one or both of a first ratio of the first N precursor flow rate to the first Ti precursor flow rate (first N/Ti flow ratio) and a second ratio of the second N precursor flow rate to the second Ti precursor flow rate (second N/Ti flow rate) are 3-100.

4. The method of claim 3, wherein the method is such that increasing one or both of the first N/Ti flow ratio and the second N/Ti flow rate decreases corresponding thicknesses of one or both of the first and second TiN thin films.

5. The method of claim 4, wherein decreasing the corresponding thicknesses decreases corresponding resistivities of the one or both of the first and second TiN thin films.

6. The method of claim 4, wherein decreasing the corresponding thicknesses increases corresponding Young's moduli of the one or both of the first and second TiN thin films.

7. The method of claim 6, wherein increasing the Young's moduli comprises increasing to a value exceeding 150 Gpa.

8. The method of claim 4, wherein decreasing the corresponding thicknesses increases corresponding hardness values of the one or both of the first and second TiN thin films.

9. The method of claim 8, wherein increasing the hardness values comprises increasing to a value exceeding 6 Gpa.

10. The method of claim 4, wherein decreasing the corresponding thicknesses decreases corresponding chlorine contents of the first and second TiN thin films.

11. The method of claim 1, wherein the ratio of the peak height or the intensity of the X-ray diffraction peak corresponding to the (111) crystal orientation of TiN to the peak height or the intensity of the X-ray diffraction peak corresponding to the (200) crystal orientation of TiN is higher for the first TiN thin film relative that for the second TiN thin film.

12. The method of claim 1, wherein the exposures to one or both of the first Ti precursor and the first N precursor during the one or more first cyclical vapor deposition cycles are at a first reactor pressure of less than about 5 torr, and wherein the exposures to one or both of the second Ti precursor and the second N precursor during the one or more second cyclical vapor deposition cycles are at a reactor pressure greater than about 5 torr.

13. A method of forming a thin film comprising titanium nitride (TiN) by a cyclical vapor deposition process, the method comprising: forming on a semiconductor substrate a first TiN thin film at a first pressure by exposing the semiconductor substrate to one or more first cyclical vapor deposition cycles each comprising an exposure to a first Ti precursor at a first Ti precursor flow rate and an exposure to a first N precursor at a first N precursor flow rate, wherein the first TiN thin film has a crystalline texture such that an X-ray spectrum of the TiN thin film has a ratio of a peak height or an intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that exceeds 0.4; and forming on the first TiN thin film a second TiN thin film at a second pressure higher than the first pressure by exposing the semiconductor substrate to one or more second cyclical vapor deposition cycles each comprising an exposure to a second Ti precursor at a second Ti precursor flow rate and an exposure to a second N precursor at a second N precursor flow rate.

14. The method of claim 13, wherein the second pressure is greater than 5 torr.

15. The method of claim 13, wherein the second TiN thin films has a preferential (111) crystalline texture such that X-ray spectrum of the second TiN thin films has a ratio of a peak height or an

intensity of an X-ray diffraction peak corresponding to a (111) crystal orientation of TiN to a peak height or an intensity of an X-ray diffraction peak corresponding to a (200) crystal orientation of TiN that is lower than the corresponding ratio of the first TiN thin film.

16. The method of claim 13, wherein at least a first ratio of the first N precursor flow rate to the first Ti precursor flow rate (first N/Ti flow ratio) is 3-100.

17. The method of claim 16, wherein a second ratio of the second N precursor flow rate to the second Ti precursor flow rate is (second N/Ti flow ratio) lower than the first N/Ti flow ratio.

18. The method of claim 17, wherein the method is such that increasing one or both of the first N/Ti flow ratio and the second N/Ti flow rate decreases corresponding thicknesses of one or both of the first and second TiN thin films.

19. The method of claim 18, wherein decreasing the corresponding thicknesses decreases corresponding resistivities of the one or both of the first and second TiN thin films.

20. The method of claim 18, wherein decreasing the corresponding thicknesses increases corresponding Young's moduli of the one or both of the first and second TiN thin films.

21. The method of claim 20, wherein increasing the Young's moduli comprises increasing to a value exceeding 150 GPa.

22. The method of claim 18, wherein decreasing the corresponding thicknesses increases corresponding hardness values of the one or both of the first and second TiN thin films.

23. The method of claim 22, wherein increasing the hardness values comprises increasing to a value exceeding 6 GPa.

24. The method of claim 18, wherein decreasing the corresponding thicknesses decreases corresponding chlorine contents of the first and second TiN thin films.

25. The method of claim 13, wherein the first pressure is less than 5 torr.
